Clément Dumas

Astra Fellow with Owain Evans, previously MATS 7 scholar with Neel Nanda.

clement[dot]dumas[at]ens-paris-saclay[dot]fr butanium.github.io Google Scholar github.com/Butanium LinkedIn

PUBLICATIONS

| lead-author papers ✦ main contributors

Training with conflicting values can induce CoT override
C. Dumas✦, J. Treutlein, J. Betley, O. Evans
LessWrong 2026

Cooperation with AIs seems to be a low-hanging fruit for better eval practices
C. Dumas✦
LessWrong 2026

Steering towards "automated grading" degrades alignment
J. Betley✦, J. Treutlein, **C. Dumas**
LessWrong 2026

Narrow Finetuning Leaves Clearly Readable Traces in Activation Differences
J. Minder✦, **C. Dumas**, S. Slocum, H. Casademunt, C. Holmes, R. West, N. Nanda
ICLR 2026; NeurIPS 2025 Mechanistic Interpretability Workshop (Spotlight)

Overcoming Sparsity Artifacts in Crosscoders to Interpret Chat-Tuning
J. Minder✦, **C. Dumas**✦, C. Juang, B. Chughtai, N. Nanda
NeurIPS 2025

What We Learned Trying to Diff Base and Chat Models (And Why It Matters)
C. Dumas✦, J. Minder✦, N. Nanda
Alignment Forum 2025

Separating Tongue from Thought: Activation Patching Reveals Language-Agnostic Concept Representations in Transformers
C. Dumas✦, C. Wendler✦, V. Veselovsky, G. Monea, R. West
ACL 2025; ICML 2024 Mechanistic Interpretability Workshop (Spotlight)

nnterp: A Standardized Interface for Mechanistic Interpretability of Transformers
C. Dumas✦
NeurIPS 2025 Mechanistic Interpretability Workshop (Poster)

Diff Mining: Logit Differences Reveal Finetuning Objectives
G. Kocher✦, R. West, **C. Dumas**, J. Minder
ICLR 2026 Workshop on Principled Design for Trustworthy AI

Pre-training Interventions, Ex Post Facto: Grafting Model Beliefs Across Checkpoints
P. Nutter✦, D. Roytburg✦, **C. Dumas**, J. Ou, S. Feng
arXiv 2026

Persona Cartography: Charting Language Model Personality Traits in Weight Space
L. Baines✦, A. G. Hawthorne✦, M. Koroliuk✦, I. Shalibashvili✦, **C. Dumas**, K. Voudouris, D. D. Africa
arXiv 2026

Believe It or Not: How Deeply do LLMs Believe Implanted Facts?
S. Slocum✦, J. Minder, **C. Dumas**, H. Sleight, R. Greenblatt, S. Marks, R. Wang
arXiv 2025

Activation Oracles: Training and Evaluating LLMs as General-Purpose Activation Explainers
A. Karvonen✦, J. Chua, **C. Dumas**, K. Fraser-Taliente, S. Kantamneni, J. Minder, E. Ong, A. Sen Sharma, D. Wen, O. Evans, S. Marks
ICML 2026 (Spotlight)

RESEARCH POSITIONS

Since Apr 2026	**Astra Fellow**, with Owain Evans. Data attribution, then model personas
Sep 2025 - Jan 2026	**SPAR mentor.** Supervised 5 junior researchers on 2 model diffing projects
May 2025 - Apr 2026	**MATS extension**, Neel Nanda's stream. 1-year scholarship to continue my model diffing research
Winter 2025	**MATS 7.0 scholar**, Neel Nanda's stream working on crosscoder and model diffing
Summer 2024	**Research intern, EPFL**, 5 months with Robert West and Chris Wendler
Summer 2023	**Research intern**, 2 months with Jobst Heitzig on Aspiration-Based Q-Learning

EDUCATION

2024-2025	Research Master MVA (Mathematics, Vision, Learning), **ENS Paris-Saclay**
2023-2024	Research Master MPRI, **ENS Paris-Saclay**
2022-2023	Double Bachelor's degree in Computer Science, **ENS Paris-Saclay**
2020-2022	"Classes préparatoires" in Mathematics, Physics and CS, Lycée Thiers

TOOLS AND PROJECTS

tinkerscope	Chat tool for exploring and comparing weird model behaviors across checkpoints
nnterp	A library that standardizes access to LLM internal activations based on NNsight
diffing-toolkit	Implements and visualizes several model diffing techniques
tiny-dashboard	A small app to visualize sparse autoencoder activations
NDIF	Part of the team building the ecosystem behind NNsight
Claude plays Nomic	Multi-agent environment where Claude instances play Nomic, a game where players vote to change the rules as they play
CodinGame	Developed a multiplayer game; top 0.5%, 3% and 7% in bot programming contests (2021-2022)

REFERENCES

Owain Evans
Director at Truthful AI, Affiliate Researcher at CHAI, UC Berkeley
owainevans.github.io

Neel Nanda
Mechanistic Interpretability Team Lead, DeepMind
www.neelnanda.io

Julian Minder
PhD Student at EPFL and Anthropic Fellow
jkminder.ch

Cameron Holmes
Senior research manager, UK AISI (previously MATS)
www.linkedin.com/in/cameronholmes1

Robert West
Associate Professor and Head of the Data Science Lab, EPFL
dlab.epfl.ch/people/west

Chris Wendler
Research Scientist, World Labs (previously EPFL)
wendlerc.github.io

Jobst Heitzig
Research Lead, Potsdam Institute for Climate Impact Research
www.pik-potsdam.de/members/heitzig

Walter Laurito
Research Engineer and Team Lead, Cadenza Lab
www.linkedin.com/in/walter-laurito-951565144

Charbel-Raphaël Segerie
Executive Director, CeSIA
crsegerie.github.io